ECON 3201
Econometrics for Economics and Finance

Assignment 1
Due Thursday October 8, 2026

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Instructions

Answer all questions. Please complete your assignment using Quarto. Submit both your Quarto file and pdf output file.

Theory Questions

Q1

Create a personal Quarto website hosted on GitHup pages. For reference, please see the slides for **website** on Nexus. Your website should include the following:

i. An "Index" or "Home" page. This page should include the following:

- Your name
- Your photo (optional)
- Your credentials
- An "About Me" section that provides a brief (one paragraph) introduction to yourself. [10 Marks]

ii. A blog page. This will be the landing site for future assignments [5 Marks]
iii. Create a blog post and upload the finished PDF for this assignment. [3 Marks]
iv. Navigation bar with "Home" and "Blog" buttons. The navigation bar is edited in the "_quarto.yml" file. [2 Marks]

> Paste your link here

Q2

Write your solutions using LaTeX. Show your work. You may use R to compute the answers.

Let $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12$ be a uniformly distributed variable with probability distribution function $f_X(x) = 1/12$.

(a) Check that $f_X(x)$ is a true probability distribution.

$$\sum_{i=1}^{12} f_X(x) = 1$$

```
f <- rep(1/12,12)
sum(f)
```

```
[1] 1
```

$\frac{1}{12} + \frac{1}{12} + \frac{1}{12} ..... + \frac{1}{12} = 1$

(b) Compute the expected value of X.

$$E[X] = \sum_{i=1}^{12} x_i f_X(x)$$

```
# Expected value of X
x <-1:12
f <- rep(1/12,12)

expected_value <- sum(x * f)
expected_value
```

```
[1] 6.5
```

$E[X] = 1 * \frac{1}{12} + 2 * \frac{1}{12} + 3 * \frac{1}{12} .....12 * \frac{1}{12} = 6.5$

(c) Compute the variance of X.

$$Var[X] = \sum_{i=1}^{12} (x_i - E[X])^2 * f_i$$

```
# Calculate variance
x <-1:12
f <-rep(1/12,12)
expected_value <-6.5
var <- sum((x-6.5)^2*f)
var
```

```
[1] 11.91667
```

$Var[X] = (1-6.5)^2 * \frac{1}{12} + (2-6.5)^2 * \frac{1}{12} + ... + (12-6.5)^2 * \frac{1}{12} = 11.91667$

Q3

Let $X \sim N(4, 5)$. Find the probability of each event:

(a) $Pr(X \leq 6)$

$$X \sim N(4,5)\,\Pr(X \leq 6)$$

```
# Let X ~ N(4, 5)
# Mean = 4, Var=5 Sd = sqrt(5)
# P(X <- 6)
pnorm(6, mean = 4, sd = sqrt(5))
```

```
[1] 0.8144533
```

(b) $Pr(X > 4)$

$$X \sim N(4,5)\,\Pr(X > 4)$$

```
# Let X ~ N(4, 5)
# Mean = 4, Var=5 Sd = sqrt(5)
# P(X <- 4)
pnorm(4, mean = 4, sd = sqrt(5))
```

```
[1] 0.5
```

(c) $Pr(-1 \leq X < 1)$

$$X \sim N(4,5)\,\Pr(-1 \leq X < 1$$

```
# Let X ~ N(4, 5)
# Mean = 4, Var=5 Sd = sqrt(5)
# P(X <- -1,1)
pnorm(1, mean = 4, sd = sqrt(5)) -
  pnorm(-1, mean = 4, sd = sqrt(5))
```

```
[1] 0.07718259
```

Q4

Let $Z \sim N(0,1)$.

(a) Compute $Pr(Z \leq 0.83)$

$$Z \sim N(0,1)\,\Pr(Z \leq 0.83)$$

```
# Let Z ~ N(0, 1)
# Mean = 0, Var=1 Sd = sqrt(1)
# P(Z <- 0.83))
pnorm(0.83, mean=4, sd=sqrt(5))
```

```
[1] 0.07814399
```

(b) Compute $Pr(-0.83 \leq Z \leq 0.83)$

$$Z \sim N(0,1)\,\Pr(-0.83 \leq Z < 0.83)$$

```
# Let Z ~ N(0,1)
# Mean = 0, Var=1 Sd = sqrt(1)
# Pz <- --0.83,0.83)
pnorm(0.83, mean=0, sd= sqrt(1)) -
        pnorm(-0.83, mean=0, sd = sqrt(1))
```

```
[1] 0.5934612
```

R Excercises

Q5

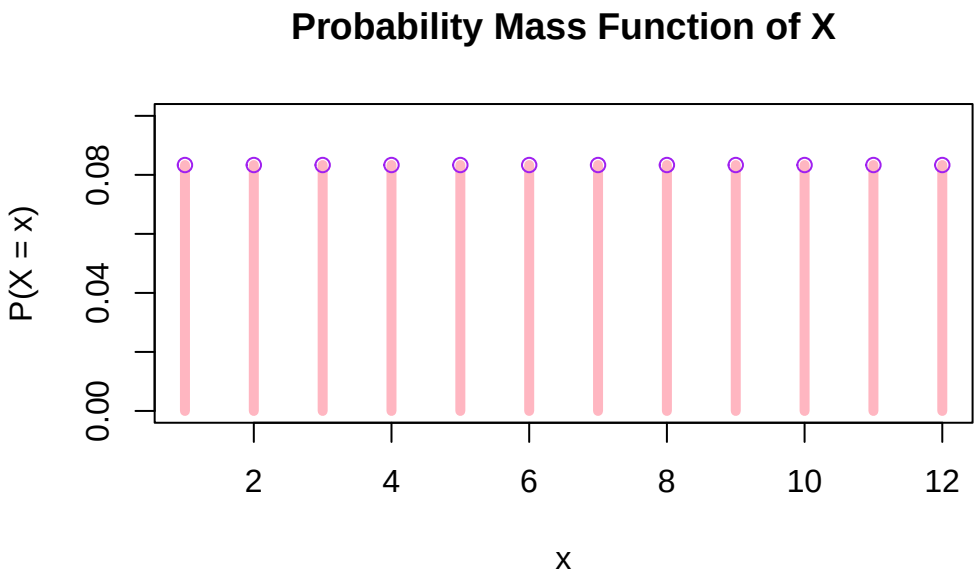
Let $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12$ be a uniformly distributed variable with probability distribution function $f_X(x) = 1/12$.

(a) Plot the probability distribution function $f_X(x)$ in R.

$$X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}\, f_X(x) = 1/12$$

```r
# Define the sample space and probabilities
x <- 1:12
f <- rep(1/12, 12)
# Plot the discrete probability distribution
plot(x, f,
     type = "h",
     lwd = 5,
     col = "lightpink",
     main = "Probability Mass Function of X",
     xlab = "x",
     ylab = "P(X = x)",
     ylim = c(0, 0.1))

points(x, f, col = "purple")
```

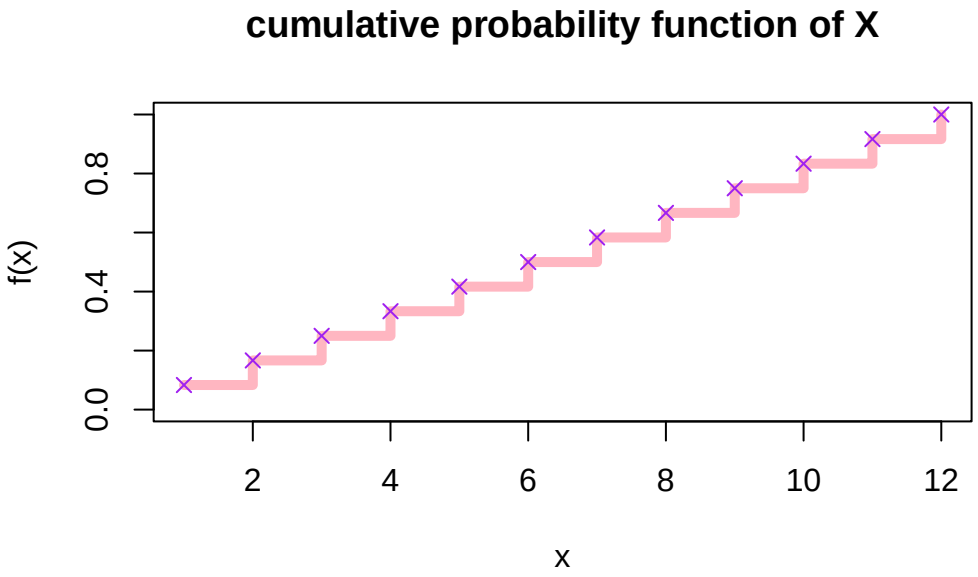


(b) Plot the cumulative probability distribution function $f_X(x) in R$.

$$F_X(x) = P(X \leq x) = \begin{cases} 0, & x < 1, \\ \frac{x}{12}, & 1 \leq x < 12, \\ 1, & x \geq 12. \end{cases}$$

```
# Define the cumulative probabilities
x <- 1:12
cdf <- cumsum(rep(1/12, 12))
# Plot the CDF
plot(x, cdf,
     type = "s",
     lwd = 5,
     col = "lightpink",
     main = "cumulative probability function of X",
     xlab = "x",
     ylab = "f(x)",
     ylim = c(0, 1))

points(x, cdf, col = "purple", pch = 4)
```



‘‘‘

(c) Compute the expected value of X in R.

```r
# Compute expected value: E[X] = sum(x * P(X=x))
x <- 1:12
pmf <- rep(1/12, 12)
expected_value <- sum(x * pmf)
expected_value
```

```
[1] 6.5
```

(d) Compute the variance of X in R.

```r
# Compute variance: Var(X) = E[X^2] - (E[X])^2
x <- 1:12
pmf <- rep(1/12, 12)
expected_value <- sum(x * pmf)

variance <- sum((x^2) * pmf) - (expected_value)^2
variance
```

```
[1] 11.91667
```

Q6

The standard normal distribution function is given by

$$f(x) = \frac{e^{-(x^2/2)}}{\sqrt{2\pi}\sigma}$$

term

(a) Define this function in R. (Note: π is coded as `pi` and e is coded as `exp`.

```r
#Define the standard Normal Distribution
f <- function(x)
  exp(-(x^2)/2) / sqrt(2*pi*1)
print(f)
```

```
function (x)
exp(-(x^2)/2)/sqrt(2 * pi * 1)
```

(b) Integrate this function over the limit $[-\infty, \infty]$ using the `integrate()` command in R. Infinity in R is written as `Inf`. To evaluate the integral, use `integrate()$value`

```
integral<-integrate(f,-Inf,Inf)$value

integral
```

```
[1] 1
```

(c) Compute the expected value of the standard normal distribution in R and confirm it is equal to 0.

```
# Expected value of a f
# Integrate from negative infinity to positive infinity
expected_value_normal<- integrate(function(x) x * f(x), -Inf, Inf)$value
expected_value_normal
```

```
[1] 0
```

(d) Compute the variance of the standard normal distribution in R and confirm it is equal to 1.

```
#Variance of f
# Integrate from negative infinity to positive infinity
variance<- integrate(function(x) x^2 * f(x), -Inf, Inf)$value
variance
```

```
[1] 1
```

(e) Compute $Pr(Z \leq 0.83)$

```
integrate(f, -Inf, 0.83)$value
```

```
[1] 0.7967306
```

(f) Compute $Pr(-0.83 \leq Z \leq 0.83)$

```
integrate(f, -0.83, 0.83)$value
```

```
[1] 0.5934612
```

(g) Compute $Pr(-0.83 \leq Z \leq 0.83)$ using the `pnorm()` command and compare your results to (f).

```
pnorm(0.83)- pnorm(-0.83)
```

```
[1] 0.5934612
```

Q7

Let X denote the prison sentence, in years, for people convicted of auto theft in a particular province in Canada. Suppose the PDF of X is given by:

$$f(X) = \frac{1}{9}x^2, \ 0 < x < 3.$$

(a) Define $f(x)$ in R.

```
#Define the PDF
#Define the standard Normal Distribution
f <- (f (x))
f <- function(x)
  x^2/9
print(f)
```

```
function (x)
x^2/9
```

(b) Integrate $f(x)$ over the interval [0,3].

```
integral<-integrate(f,0,3)$value

integral
```

```
[1] 1
```

(c) Compute the expected prison sentence.

```
# Expected value of a f
# Integrate from 0 to 3
expected_value_normal<- integrate(function(x) x*  x^2/9, 0, 3)$value
expected_value_normal
```

```
[1] 2.25
```

Q8

Let X follow a Student t distribution with M degrees of freedom. Then

$$X = \frac{Z}{\sqrt{W/M}} \sim t_M,$$

where Z is a standard normal variable and W is a chi-squared variable with M degrees of freedom.

(a) On the same figure plot the following:

(i) Plot the standard normal probability distribution using `curve()`. Specify the range as [-5,5] using `xlim`. Name the `ylab` "Density". The main title should be "Densities of t Distribution".

(ii) Plot the Student t distribution with 5 degrees of freedom using `curve()` and `dt()`. Specify the range as [-5,5] using `xlim`. Choose a different colour, i.e. `col=2`. To learn more about `dt`, type `?dt` in your console.

(iii) Plot the Student t distribution with 15 degrees of freedom using `curve()` and `dt()`. Specify the range as [-5,5] using `xlim`. Choose a different colour, i.e. `col=3`. To learn more about `dt`, type `?dt` in your console.

(iv) Plot the Student t distribution with 25 degrees of freedom using `curve()` and `dt()`. Specify the range as [-5,5] using `xlim`. Choose a different colour, i.e. `col=4`. To learn more about `dt`, type `?dt` in your console.

(v) Include the following legend in you plot:
`legend("topright", c("N(0, 1)", "M=2", "M=4", "M=25"), col = 1:4, lty = c(1, 1, 1, 1))`

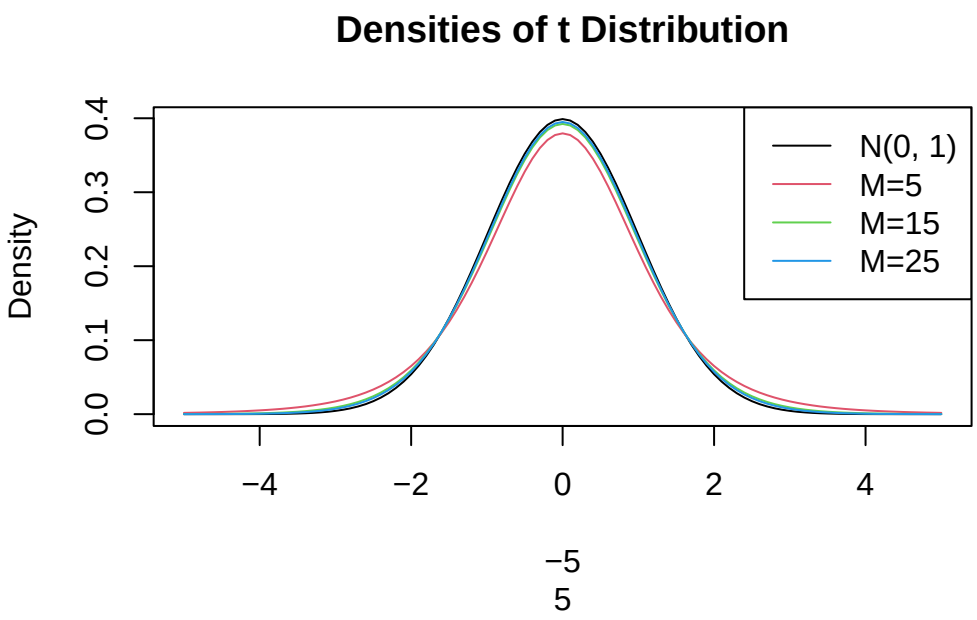
```r
# (a) Plot the standard normal and Student t distributions

# (i) Standard Normal Distribution
curve(dnorm(x), from = -5, to = 5,
xlab = c(-5,5), ylab = "Density",
main = "Densities of t Distribution",
col = "1")

# (ii) Student T distribution, df=5
curve(dt(x, df =5),
            from = 5, to = -5,
            xlab = c(-5,5), ylab = "Density",
            col = "2",add = TRUE)
?dt

# (iii) Student T distribution df=15
curve(dt(x, df = 15),
            from = 5, to = -5,
            xlab = c(-5,5), ylab = "Density",
            col = "3", add = TRUE)
?dt

#(iv) Student T distribution df=25
curve(dt(x, df =25),
            from = 5, to = -5,
            xlab = c(-5,5), ylab = "Density",
            col = "4",add = TRUE)
?dt
#(v)
legend("topright",
       c("N(0, 1)", "M=5", "M=15", "M=25"),
       col = 1:4,
       lty = c(1, 1, 1, 1))
```



(b) What do you observe as the degrees of freedom increase?

As the degrees of freedom increase, the Student t distribution approaches the standard normal distribution. It also confirms the central limit theorm, that is as n becomes larger, the actual function becomes better appoximated by the normal distribution

#quarto render econ_3201_a1.qmd –to pdf